Azure Migration Security Report

Introduction

As a security engineer, the primary goal of this migration plan is to protect the company's data and resources in the cloud. It advices in securing user accounts, applications, data, virtual private networks, and virtual machines against unauthorized access and vulnerabilities using Microsoft Azure services.

User Accounts and Identity Management

The foundation of security starts with managing user identities. Azure Active Directory (Azure AD) is used to handle authentication and access control. Once the Azure AD tenant is set up and user accounts are created, implement Role-Based Access Control. This ensures users only have the permissions they need. For example, some users will have read-only access to critical resources or general data, while others will be assigned roles with full read and write permissions. To further protect accounts, require Multi-Factor Authentication for everyone. Set up Conditional Access policies to restrict access based on user location, device safety, and risk levels. Regular access reviews will be conducted to confirm that user permissions remain appropriate over time.

Application Security

To securely deploy and manage applications, use Azure App Service. This service allows configuring secure communication channels using SSL and TLS encryption. Utilize deployment slots for staging and testing environments to prevent any risks to the live production systems. Sensitive information, such as API keys and connection strings, will never be left in the code. Instead, they will be stored safely in Azure Key Vault and program applications to retrieve these secrets securely.

Data Protection and Monitoring

All company data will be stored securely using Azure Storage. Guarantee that data is encrypted while at rest using Azure Storage Service Encryption. To control who and what can reach this stored data, apply strict firewall rules and use virtual network service endpoints. Furthermore, rely on Azure Security Center to constantly monitor the resources for new threats and vulnerabilities. Actively review and implement the security alerts and recommendations it provides.

Network and Remote Access

Connecting on-premises network to the Azure cloud requires a secure bridge. Set up an Azure VPN Gateway to ensure all connections between the local network and Azure resources are heavily encrypted. When administrators need to manage virtual machines directly, they will use Azure Bastion. This service provides a safe and smooth

remote desktop and SSH connection directly through the Azure portal, meaning the virtual machines will never be dangerously exposed to the public internet.

Virtual Machine Security

As virtual machines are deployed running Windows operating systems, implement strict network controls. Use Azure Firewall to monitor and filter all traffic entering and leaving the virtual network. Additionally, configure Network Security Groups to define specific rules that allow or block traffic to individual virtual machines. Activate Microsoft Defender for Cloud to provide advanced threat protection and continuous vulnerability management for all virtual instances.

Conclusion

By moving away from basic lists and implementing these structured security measures, the company ensures a highly secure Azure environment. This comprehensive approach utilizes Azure's built-in services to maintain strong defenses for the network, applications, and data.

This report was conducted by Jenna Hemtman on 09/2024 to complete study unit "Azure Security Technologies" from Microsoft Skills for Jobs, organized in collaboration with Kajaanin Ammattikorkeakoulu.